RESEARCH REPORT · VITKI

Fully tested finding

Uppland's runestones differ across the old folkland borders – twice as much as across random borders through the same districts

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Timeline of the study: The map layer of the old lands was built and the folkland border for at was tested under a locked plan 1 November 2026 · The hundreds' folklands were checked against the six-year tithe of 1314, the border scan was run, and the random-partition test and the addendum plan for the sensitivity analyses were locked and run 2 November 2026 · The land assignment was corrected and every test was rerun 5 November 2026 · The preposition feature was corrected and every test was rerun 7 November 2026

ABSTRACT

Background and aim. In the Middle Ages Uppland was not one province but three folklands – Tiundaland, Attundaland and Fjädrundaland – and the coastal land Roden, with a common law only from 1296. How old the folklands are is not known: they are first mentioned around 1120, and it has been suggested that the division goes back to the late tenth century. The runestones were raised precisely in the eleventh century. If the folkland borders already separated people then, the stones should differ more across those borders than across other borders through the same districts.

Material. 873 Viking-age runestones in Uppland with known location, in 31 hundreds and ship districts with their folkland according to the six-year tithe of 1314 and, for the islands of Lake Mälaren, *Det medeltida Sverige*; 25 features of language, formulas, family, carving and stone. Before publication the land assignment was corrected: 39 stones had dropped out because their parishes had no folkland, and 32 of them are now included. Then the coding of the preposition *at* was corrected.

Method. For each feature we measured how much block membership explains, with time held constant (logistic regression, likelihood-ratio test), and the sum over the features was compared with the same sum for 999 random divisions of the same hundreds into four contiguous blocks of the same size. The plan was locked before the run.

Results. The real folklands give a sum of 347; the random divisions give a median of 178, and the top twentieth begins at 248. Only 2 of 999 reach as high ($p = 0.003$; Holm-corrected 0.006), in both halves none ($p = 0.001$), and the test holds in all four pre-registered sensitivity analyses (p 0.002–0.007). Before

the two corrections: 360, and none reached it ($p = 0.001$). Småland's small lands differ no more than random blocks ($p = 0.267$).

Conclusion. The folkland borders separate the runestones about twice as much as a typical random border through the same districts. This suggests that the division, or something that preceded it, existed already when the stones were raised, but the test does not decide what set them apart – language, custom or workshops.

Keywords: *runestones, Viking Age, Uppland, folkland, hundred, regional variation, permutation test, logistic regression*

HIGHLIGHTS

- Uppland's folklands separate the runestones more than almost every other way of dividing the same districts into equally large, contiguous blocks; the borders are not just any borders.
- The differences show both in how the stones were written (r for ʀ, the monophthong) and in the custom (the cross, several raisers). The preposition *at* also differs, but the sharp border for *at* between Tiundaland and Attundaland that we first saw does not hold when the word is counted correctly.
- That the division, or the districts it was built on, existed already when the stones were raised is an interpretation the result supports but does not prove; why the borders separated – language, custom or workshops – the test does not decide.
- Småland's small lands show nothing similar.

1. Introduction

That Uppland's runestones vary within the province is known. Palm (1992) showed that the preposition *at* is more common in Tiundaland than in Attundaland and drew his study areas along the folkland borders. Larsson (2002, pp. 168–170) asked why Uppland, Gotland and Västergötland more often write ʀ with the r-rune and held that for Uppland it is hardly the geographical position, rather chronological and partly individual factors. Stille (1999) studied the carvers of a single folkland, Fjädrundaland. Peterson (1996) showed that *aft/æft* is the oldest form and that *at* and *æftiR* mean the same; in formulas with both words *at* nearly always comes after *æftiR* (pp. 246 f.). She tests no border within Uppland. On the age of the folklands, Mats G. Larsson (1993, p. 114) has suggested that both the levy and the folkland division were organised by Erik the Victorious in the later tenth century; the folklands are first named in a Latin text of about 1120, and two of them in an account that may go back to the mid-eleventh century (pp. 106 f.).

What is missing is a test of the borders themselves. That two areas differ says nothing yet about the border between them: neighbouring districts resemble each other, and almost any division of a province produces differences between its parts. The question is therefore whether the folkland borders separate the stones more than other borders, drawn through the same districts, do. It is tested here with a permutation test in which the real folklands are compared with random divisions of the same hundreds.

2. Pre-registration

The plan (*New patterns with the Viking-age division*, step 1) was locked on 2 November 2026 with SHA-256 054c87d521930002..., before the test was run. The hypothesis, translated in full: "If the folklands were merely arbitrary blocks of neighbouring districts, any blocks would differ as much, since neighbouring stones resemble each other." The decision rule: the folklands are special if their sum lies above the 95th percentile of the random divisions ($p = (1 + \text{number} \geq \text{real}) / 1000 < 0.05$), and the same rank is reported in both halves.

One deviation was locked before any result had been computed (SHA-256 f0a42169ae5133a7...): the neighbour rule had to be widened, see 4.4. The sensitivity analyses were locked in a separate addendum after the main test but before they were run (SHA-256 e31f2fd146b4c631...). The land assignment was corrected after the review before publication, and then the coding of the preposition *at*; both times everything was rerun, after the results had been seen (4.4).

Seen before locking: that *at* changes sharply at the border between Tiundaland and Attundaland (its own locked plan the day before, SHA-256 48346775da5a5f30...; with the coding that was later corrected, see 5.4), and that a border scan (its own plan, SHA-256 7d22ff404b2c88ef...) had found sharp borders, all between the folklands. The main test is therefore not a blind discovery: it tests whether a pattern seen at single borders holds for the division as a whole, against an explicit null hypothesis. Confirmatory: step 1 for Uppland. Secondary: step 1 for Småland. Follow-up under its own locked plan: the sensitivity analyses. Exploratory: the test without the rock type and which features carry the difference.

3. Material

The runestones come from the Scandinavian Runic-text Database (Rundata), 2014 version with RUNDATA.xls updated 2018, frozen for the analysis on 4 October 2026. The selection: Viking-age runestones in Uppland whose parish can be linked to a hundred in the place-name register of the Institute for Language and Folklore (the same parish name within 25 km, otherwise the nearest parish within 15 km) and whose hundred belongs to a folkland: 873 stones in 31 hundreds and ship districts. The distribution: Attundaland 8 hundreds and 366 stones, Tiundaland 11 and 320, Fjädrundaland 5 and 128, Roden 7 and 59.

The folkland of each hundred follows the six-year tithe of 1314 (SDHK 2593), which lists Attundaland's eight hundreds and Fjädrundaland's five as here; Tiundaland there also includes Gästrikland and Roden. Roden is treated here as a block of its own. Its extent and status are disputed already in the medieval material (Dahlbäck et al. 1972).

The land correction before publication. In the first run 39 stones from Uppland dropped out unnoticed, because their parishes had no folkland: 24 in Färingö tingslag, 6 on Adelsö, 5 in Ekerö and four in Stockholm. According to *Det medeltida Sverige* (Ferm et al. 1992, pp. 19, 44, 114), Färingö tingslag was "the other half" of the hundred of Bro in Attundaland, Adelsö belonged to the hundred of Trögd in Fjädrundaland until 1540, and Ekerö to the hundred of Svartlösa in Södermanland; they are counted so here. U 419 and U 871 at Skansen are counted to their parishes, while U 53 and U 54

in the town of Stockholm, outside the hundreds, are not included. Spånga and Bromma belong to the hundred of Sollentuna (p. 237), so U 77 moved from Roden to Attundaland. Before the correction 841 stones in 32 units were included, after it 873 in 31.

The 25 features are those of the plan's 27 that have at least ten stones with and ten without the feature: the cross, a Christian marker, prayer and prayer for the soul, journey and death abroad, praise, bridge and road, a woman among the raisers, a woman as sole raiser, several raisers, stones for a child, parent, sibling and spouse, *drengr* and *þegn*, the carver's signature, the rock type granite, dotted runes, monophthong spelling, r for ʀ, the o-rune, the preposition *at* (counted before a name, a kinship word or *sik*/*sinn*, see 4.4) and the word *þennsa*. The plan's two other features, a pagan marker and the word *kumbl*, fell out because too few stones have them. Time is held constant with five strata by the midpoint of the style group (before 1010, 1010–1040, 1040–1070, after 1070, undated). The features are coded from Rundata's transliterations, normalisations and translations with the project's rules; the coding has not been hand-checked for this study.

4. Method

4.1 The model. For each feature a logistic regression with the time stratum and the block membership is estimated, and the likelihood-ratio test of the block term gives a χ^2 value: how much better the stones' features are predicted when one knows which block they stand in. The measure is the sum of χ^2 over all 25 features. The second measure is the number of features with $p < 0.05$.

4.2 The random divisions and the halves. 999 random divisions are built from the same 31 hundreds into four blocks with as many hundreds as the real folklands (8, 11, 5 and 7). Each block grows from a randomly chosen starting hundred to a randomly chosen neighbour until it has its size; a division that cannot be completed is redrawn. The halves: Uppland's parishes are split alternately by mean dating, and the test is repeated in each half with the same 999 divisions.

4.3 The sensitivity analyses. The same test on four selections according to the addendum: only stones in their original place; without all stones by the five most prolific carvers (Öpir, Fot, Balle, Åsmund, Gunnar, certain and uncertain attributions); only dated stones; without lost stones. Each step holds if $p < 0.05$.

4.4 Deviations from the plan. (1) The plan's neighbour rule (parishes within 10 km) left the ship district of Värmdö and the hundred of Våla without neighbours, so that no contiguous divisions could be built; before any result it was widened to 20 km, and a hundred without neighbours gets its nearest as neighbour (locked deviation). (2) The test without the rock type granite was run after the main result and lies outside the plan; it is reported as exploratory. (3) Which features carry the difference was computed after the result and is descriptive. (4) The land assignment was corrected on 5 November 2026, after the results had been seen, and everything was rerun with the same code and random seed: the medieval hundred of eleven parishes after *Det medeltida Sverige* (section 3), Rundata's "(nu …)" ("now …") after the parish name is dropped, and a stone moved far from its parish is linked to it. The method is unchanged, and the first run's results are kept. (5) The preposition feature was corrected on 7 November 2026, also after the results had been seen: *at* was counted only before a name, so that *at* as the second preposition was missed ("… ok Fastlaug at bonda sinn",

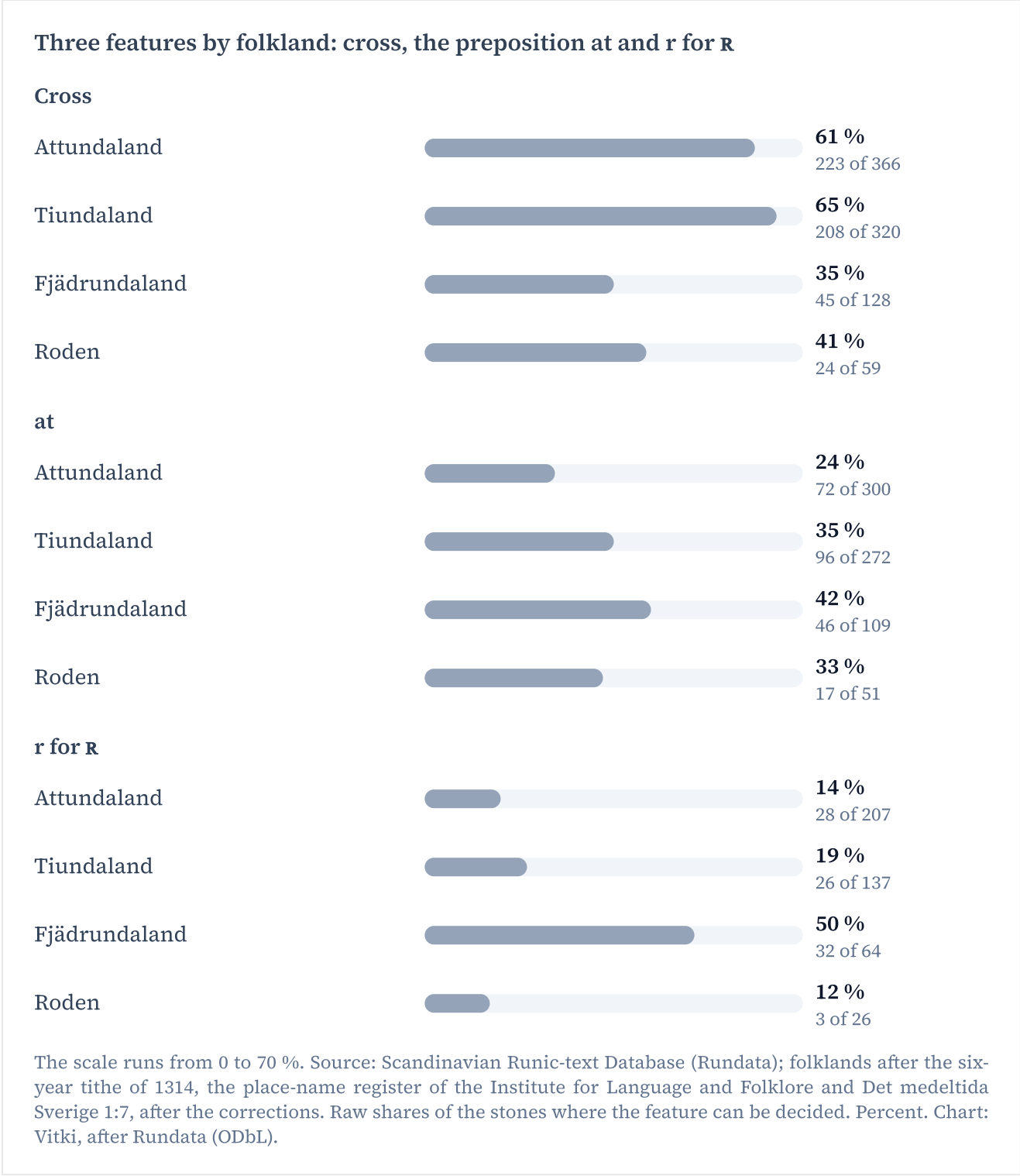
U 461). The feature changed on 206 of 1,739 stones, and everything was rerun in the same way. No other deviations.

5. Results

5.1 The main test. The real folklands give a sum of 347. The 999 random divisions give a median of 178, and the 95th percentile lies at 248. Two random divisions reach as high: p = 0.003, and Holm-corrected over the plan's two tests 0.006. The number of differing features is 15 against a median of 7 (p = 0.013). In the halves: 238 against a median of 132 and 207 against 121; no random division reaches the folklands in either (p = 0.001 in both). Before the land correction: 360 against a median of 183, no random division reached it (p = 0.001), and the halves 237 and 217 (p = 0.001 in both).

In plain words: the folkland borders separate the language, custom and execution of the rune-stones about twice as much as a typical random border drawn through the same hundreds, and of 999 random ways of dividing Uppland into four equally large, contiguous parts, only two reached the level of the real folklands. If the folklands were merely arbitrary blocks, a value this high would arise in about three cases in a thousand.

Småland's small lands, the secondary test: 18 hundreds (before the land correction 19), 103 stones and 14 features; the sum 193 against a median of 184 (p = 0.267; before the land correction 0.145). The small lands separate the stones no more than random blocks do.



5.2 Which features (descriptive). The five features that contribute most are the cross, the rock type granite, monophthong spelling, r for ʀ and several raisers; the preposition *at* comes eighth. The cross is on six stones in ten in Attundaland and Tiundaland but on one in three in Fjädrundaland and four in ten in Roden; r for ʀ on half the stones in Fjädrundaland against one in seven in Attundaland; *at* on about one stone in four in Attundaland (one in seven before the preposition was counted correctly) against four in ten in Fjädrundaland.

This is what the difference looks like on the stones. The same carver, Öpir, wrote *at* on 23 of 40 stones in Tiundaland but on 9 of 25 in Attundaland, and the same word, *þæiR* 'they', is spelled with

the r-rune in Fjädrundaland but with the yr-rune in Tiundaland (the runes are rendered with the usual younger-futhark forms after Rundata's transliteration, not with the stones' exact forms):
- Tiundaland, U 1177 at Harbo, carved by Öpir: ᚴᛁᛏᚢ·ᚱᛁᛋᛏ·ᛋᛏᛁᚾ·ᚾᛏ·ᛏᚢᛒᚾ – *litu risa stin at tuba*, 'had the stone raised in memory of Tobbe'.
- Attundaland, U 181 at Össeby-Garn, carved by Öpir: ᛁᚠᛏᛁᚼ·ᚨᛋᛏ·ᚠᚨᚦᚢᚱ·ᛋᛁᚾ – *iftiR est faþur sin*, 'in memory of Äst, their father'.
- Fjädrundaland, U 749 at Husby-Sjutolft, attributed to Balle: ᚦᚨᛁᚱ·ᚴᛁᛏᚢ·ᚱᚨᛋᚨ·ᛋᛏᚨᚾ – *þair litu rasa stan*, 'they had the stone raised', with the ʀ of *þæiR* written with the r-rune ᚱ.
- Tiundaland, U 974 at Vaksala, attributed to Öpir: ᚦᚨᛁᚼ – *þaiR*, the same word with the yr-rune ᚼ.

5.3 The sensitivity analyses and the calibration. All four pre-registered steps hold, as before the land correction (then p between 0.002 and 0.003):
- Only stones in their original place (513 stones): 234 against a median of 136, p = 0.007.
- Without the stones of the five most prolific carvers (559): 229 against 134, p = 0.002.
- Only dated stones (529): 263 against 142, p = 0.004.
- Without lost stones (734): 295 against 165, p = 0.005.
- Exploratory, without the rock type granite (24 features): 308 against 147, p = 0.005; in the halves p = 0.002 and 0.003 (before the land correction p = 0.001). The difference is thus not carried by the rock type granite.

The calibration: the measure is compared with the random divisions, not with zero; contiguous random blocks already give a median of 178, because neighbouring districts resemble each other. With 999 divisions p cannot be lower than 0.001 (Phipson & Smyth 2010). The model has not been checked in other ways. That 32 stones came in through the land correction changed the sum from 360 to 363 and p from 0.001 to 0.003; the correction of the preposition changed the sum to 347 with p still 0.003. Neither correction changed the outcome.

5.4 Follow-up: the borders one by one. A border scan (its own locked plan) tested every pair of neighbouring regions with at least 40 stones and asked whether the difference remains among the stones within ten kilometres of the border. 7 combinations of 6 features are sharp at a border and hold in both halves: the rock type granite between Tiundaland and Attundaland; the cross, several raisers, praise and r for ʀ between Tiundaland and Fjädrundaland; several raisers between Attundaland and Fjädrundaland (only 16 stones); and *at* between Södermanland and Fjädrundaland (13 of 13 stones on the Södermanland side against 3 of 10 on Fjädrundaland's).

at between Tiundaland and Attundaland is still sharp (49 of 131 stones on Tiundaland's side against 15 of 78), but since the preposition is counted correctly it does not hold in both halves (p = 0.009 and 0.218), nor in the folkland plan (p = 0.186 and 0.009). With the carver as a random effect the odds ratio for *at* on Tiundaland's side is 2.7 (before the correction 5.9), but without the hundred of Seminghundra it is not enough (p = 0.062). That border therefore no longer counts as tested.

Between Attundaland and Fjädrundaland on the islands of Lake Mälaren there are now stones on both sides (3–14 per feature, before at most 2), but for most features too few; there *at* is not sharp (1 of 4 against 7 of 9, p = 0.993).

6. Discussion

The result says that the folkland borders are not just any borders: they separate the stones more than almost all contiguous random blocks of the same size. Since the stones are mainly from the eleventh century and the folklands are known only from around 1120, this suggests that the division – or the district communities it was built on – existed already when the stones were raised. This agrees with Larsson's (1993) proposal of an organisation from the late tenth century, but the test cannot tell an administrative division from older districts that the division later followed.

Against earlier research: Palm (1992) already saw the difference in *at* between Tiundaland and Attundaland, and Larsson (2002) that Uppland is among the areas where ʀ is more often written with the r-rune. What is new here is that the borders are tested as a whole, against a null hypothesis that takes into account that neighbouring districts resemble each other, and that they hold.

The weaknesses in order of importance: (1) The division is known from the fourteenth century. The hundreds' folklands rest on the six-year tithe of 1314 and the hundreds' borders on later registers; the hundreds may have looked different in the eleventh century. (2) The land assignment and the preposition were corrected after the results had been seen; neither correction changed the outcome, but single parishes are uncertain (Lovö was a ship district of its own in 1409 and is counted here to the hundred of Bro). (3) Roden is disputed and is treated here as a block of its own. (4) The features are related (the cross and Christian formulas, several raisers and the relations), so the sum of χ^2 counts such features more than once; the test is still valid since the null distribution rests on the same features, but the measure is not a count of independent differences. (5) The random blocks have as many hundreds as the folklands, not as many stones. (6) The carvers: without the five most prolific the test holds, but smaller workshops may still carry part of the difference; the test does not separate land from workshop. (7) Småland's negative result may be due to too few stones. (8) The features are coded by rules, not by hand; the preposition *at* was miscoded until Peterson (1996) was read, and other features may have errors of the same kind.

The finding falls if other contiguous divisions of Uppland that follow natural borders, such as watercourses or forest belts, separate the stones as much as the folklands; or if an independent material, such as burial customs or place names, does not show the same borders.

7. Conclusion

Uppland's folklands separate the runestones about twice as much as a typical random division of the same districts, and only two of 999 random divisions reach as far. This suggests that the division, or the districts it was built on, existed already in the eleventh century.

Data and code

The analysis was done in Vitki 3.1.0, an open research tool for runestones (vitkirunor.se). Code state in https://github.com/Precatio/runes: the first run in commit 8d24b20 (2–3 November 2026),

the rerun after the land correction in commit 7455a18 (5 November 2026) and the rerun after the preposition was corrected in commit ddcbe70 (7 November 2026). The code and result files are there under GPL-3.0.

- Data: Rundata (2014 version with RUNDATA.xls updated 2018), frozen 2026-10-04. The feature table `utdata/upptackt_v3/drag.csv` on the material's version (corpus fingerprint) `e83202e01a5e6d27`, computed 2026-10-29, with the column `prep_at` recomputed 2026-11-07 from `510ebfb3acc9a229` (the original in `drag_v1.csv`); moved and lost stones for the sensitivity analyses from `7b9061c9a1c12c73` (2026-11-02). ISOF's parishes and hundreds retrieved 2026-11-02 (PlaceNameService, CC BY). The land assignment `web/src/lib/historiska_land.json` (the corrections under `parish_medieval`), `data/geo/socknar_land.csv` and `web/public/geo/sten_land.json` in corrected form 2026-11-05; before the correction commit 461dca4; the stones that changed land in `utdata/folkland/landrattelse.json`.
- The plans: `analysplaner/2026-11-02-gamla-granser-nya-monster.md`, the deviation `analysplaner/2026-11-02-gamla-granser-nya-monster-avvikelse.md`, the addendum `analysplaner/2026-11-02-folkland-slumpindelning-trappa.md`, with checksums beside them.
- The main test: `utdata/gamla_granser/monster.py` → `monster.json`; the sensitivity analyses `monster_trappa.py` → `monster_trappa.json`; the test without granite `monster_kanslighet.py`; the features `monster_drag.py`; the first run in `*_v1.json` and the run before the preposition was corrected in `*_v2.json` beside them. The border scan: `utdata/granser/gransskannern.py`; *at* with the carvers and the ladder: `utdata/folkland/at_ristare.R` and `at_trappa.R`; the folkland plan `utdata/folkland/folkland.py`; the reading of Peterson and the check of the coding `utdata/folkland/peterson1996_lasning.md`.
- The regions: `src/regions.py`, `scripts/assign_historiska_land.py` (the ISOF place-name register as first source, Det medeltida Sverige for the corrected parishes), `scripts/assign_sten_land.py`, the check against the six-year tithe `utdata/geografi/sexarsgard_1314.py`.
- The report is built by `utdata/blogg/build/build_rapport_folkland.py`, which reads every number from the result files and stops if anything does not match. The template: `utdata/rapporter/MALL.md`, version 4. Glossary: https://vitkirunor.se/en/forskningsrapporter/begrepp.
- In Vitki the folklands are shown on the map (the border layer Small lands) and in the statistics by the old borders. Told for a general audience in the series The old lands.

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